

1 Phase 6: Express.js (Backend Development)

2 Objective

Yo phase ma timi React frontend bata backend development tira janxau. Backend ko main kaam user bata request receive garne, database sanga communicate garne, business logic execute garne ra response pathaune ho.

Learning Outcome:

- Express.js ko basic bujhne
 - REST API banauna sakne
 - Authentication & Authorization implement garna sakne
 - Secure backend banauna sakne
 - API-only project develop garna sakne
-

3 1. Routing

4 What is Routing?

Routing vaneko **URL request lai specific function sanga connect garne process** ho.

Example:

```
GET    /users
POST   /users
PUT    /users/:id
DELETE /users/:id
```

Purpose

- API endpoints define garna
 - Different requests handle garna
-

5 2. Middleware

6 What is Middleware?

Middleware vaneko request ra response ko bich ma chalne function ho.

Flow

```
Request
  ↓
Middleware
  ↓
Route
  ↓
Response
```

Common Uses

- Authentication
 - Logging
 - Validation
 - Error Checking
-

7 3. Controllers

Controller ma application ko **business logic** lekhinxa.

Flow

```
Route
  ↓
Controller
  ↓
Database
  ↓
Response
```

Controller ko kaam

- User Create
 - User Update
 - Delete
 - Login
 - Register
-

8 4. Services

Service layer ma reusable business logic rakhinxa.

Flow

```
Controller
↓
Service
↓
Database
```

Example

- Password Hash
- Email Send
- JWT Generate
- Payment Logic

9 5. MVC Architecture

MVC = **Model** + **View** + **Controller**

Express API ma mostly

```
Route
↓
Controller
↓
Model
↓
Database
```

Large projects ma

```
Route
↓
Controller
↓
Service
↓
Model
↓
Database
```

10 6. REST API

REST API vaneko standard way ma API banaune method ho.

CRUD Operations

GET	Read
POST	Create
PUT	Update
DELETE	Delete

Example

```
GET /products
POST /products
PUT /products/5
DELETE /products/5
```

11 7. Validation

Validation le user bata aako data correct xa ki xaina check garxa.

Example

```
Email required
Password minimum 8 characters
Name cannot be empty
```

Popular Libraries

- Express Validator
 - Joi
 - Zod
-

12 8. Cookies

Cookie browser ma save hune sano data ho.

Uses

- Login Session

- Remember User
 - Authentication
-

13 9. CORS

Frontend

`localhost:5173`

Backend

`localhost:5000`

Browser le different origin block garna sakxa.

CORS le frontend lai backend access garna permission dinxa.

14 10. Helmet

Helmet security middleware ho.

Protects against

- XSS
- Clickjacking
- MIME Sniffing

Almost every production Express app ma use hunxa.

15 11. Morgan

Morgan logging middleware ho.

Console ma request information dekhauxa.

Example

`GET /users`

200 OK

35 ms

Useful for debugging.

16 12. Compression

Compression response size ghatauxa.

Without Compression
500 KB

↓

With Compression
120 KB

Benefits

- Faster API
 - Less Bandwidth
 - Better Performance
-

17 13. Rate Limiter

Ek user le dherai request pathayo vane server overload huna sakxa.

Rate Limiter le

Maximum

100 requests/minute

matra allow garxa.

Benefits

- Prevent Spam
- Prevent Brute Force
- Protect Server

18 14. Error Handling

Application crash huna nadina centralized error handling garinx.

Example

404 Not Found

401 Unauthorized

403 Forbidden

500 Internal Server Error

19 15. Authentication

Authentication = **Who are you?**

Example

Email

Password

↓

Verify

↓

JWT Token

User genuine ho ki haina verify garxa.

20 16. Authorization

Authorization = **What are you allowed to do?**

Example

Admin

✓ Delete User

✓ Add Product

User

✗ Delete User

✓ View Product

21 Recommended Folder Structure

```
project/  
src/  
├─ routes/  
├─ controllers/  
├─ services/  
├─ models/  
├─ middleware/  
├─ validators/  
├─ config/  
├─ utils/  
├─ app.js  
└─ server.js  
.  
env  
package.json
```

22 Final Project (API Only)

Build **Inventory Management API**

Features

- User Registration

- Login (JWT)
- Authentication
- Authorization
- Products CRUD
- Categories CRUD
- Validation
- Error Handling
- CORS
- Helmet
- Morgan
- Compression
- Rate Limiting

Testing Tools

- Postman
 - Thunder Client
 - Insomnia
-

23 Learning Order

1. Express.js Introduction
2. Routing
3. Middleware
4. Controllers
5. Services
6. MVC
7. REST API
8. Validation
9. Cookies
10. CORS
11. Helmet
12. Morgan
13. Compression
14. Rate Limiter
15. Error Handling
16. Authentication
17. Authorization
18. Build API-only Project